Islamic University of Technology

Department of Computer Science and Engineering

Cross-Lingual Information Retrieval (CLIR) System

Final Report

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Submitted By

Ashfak Azad Nafi
210041245
Sayad Ibne Azad
210041251
Rashikh Ahmad
210041255

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Abstract

This project presents a complete Cross-Lingual Information Retrieval (CLIR) system for retrieving relevant news articles between English and Bangla. The objective is to enable users to submit queries in one language and retrieve relevant documents in the same or another language, addressing the linguistic gap in multilingual information access. The system was built using a modular architecture consisting of dataset construction, query processing, retrieval modeling, and evaluation components.

A dataset of approximately 973 real-world news articles (330 English and 643 Bangla) was collected from major Bangladeshi news portals. Each document was processed and stored in structured JSON format with metadata including title, URL, source, language, and category.

The query processing module performs automatic language detection, normalization, named entity mapping, cross-lingual translation, and query expansion to improve retrieval robustness. Three retrieval approaches were implemented and compared: (1) lexical retrieval using BM25, (2) semantic retrieval using multilingual sentence embeddings, and (3) a hybrid model combining lexical, semantic, and fuzzy matching scores through normalized weighted fusion.

The system was evaluated on 20 manually labeled cross-lingual queries using standard Information Retrieval metrics: Precision@10, Recall@50, Mean Reciprocal Rank (MRR), and nDCG@10. Experimental results demonstrate that the hybrid model significantly outperforms individual lexical and semantic models, achieving Precision@10 of 0.27, Recall@50 of 1.00, MRR of 0.75, and nDCG@10 of 0.78, while maintaining an average retrieval time of approximately 120 milliseconds per query.

The findings confirm that combining lexical and semantic representations is highly effective for cross-lingual retrieval tasks. This project highlights the importance of multilingual embeddings, query expansion, and hybrid ranking strategies in improving retrieval accuracy across languages.

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1 Introduction

In today's digital world, vast amounts of information are generated and published daily across multiple languages. News portals, blogs, government websites, and research repositories continuously produce content in both global and regional languages. However, accessing relevant information across language barriers remains a significant challenge. Traditional Information Retrieval (IR) systems are primarily designed for monolingual environments, meaning that users can effectively retrieve documents only if the query and documents are written in the same language. This limitation restricts access to valuable multilingual information.

Cross-Lingual Information Retrieval (CLIR) addresses this problem by enabling users to submit a query in one language and retrieve relevant documents written in another language. CLIR plays a crucial role in multilingual societies, international research environments, and global information access systems. For countries like Bangladesh, where both Bangla and English are widely used in news reporting and official communication, cross-lingual retrieval becomes particularly important.

Despite advances in multilingual Natural Language Processing (NLP), cross-lingual retrieval remains a challenging task. The primary difficulties include:

- Differences in vocabulary and grammar across languages
- Named entity variations ("Dhaka" vs "ঢাকা")
- Translation ambiguities
- Limited high-quality bilingual dictionaries
- Morphological and syntactic variations
- Semantic mismatches between translated queries and documents

Traditional lexical retrieval models such as TF-IDF or BM25 rely heavily on exact term matching. These models perform well in monolingual retrieval but struggle in cross-lingual settings because translated queries may not perfectly match document terms. On the other hand, modern neural embedding models generate semantic representations of text, enabling retrieval based on meaning rather than exact word overlap. However, purely embedding-based systems may sometimes ignore important keyword-level signals.

To address these limitations, this project proposes a **Hybrid Cross-Lingual Information Retrieval System** that combines lexical and semantic retrieval techniques. The system supports both English and Bangla queries and retrieves relevant news articles from a multilingual dataset. It integrates multiple components including query normalization, language detection, translation, named entity mapping, lexical scoring using BM25, semantic similarity scoring using multilingual sentence embeddings, and weighted hybrid ranking.

The dataset used in this project consists of approximately 973 real-world news articles collected from prominent Bangladeshi news portals. The dataset includes 330 English documents and 643 Bangla documents. Each document contains structured metadata such as title, URL, source, language, and category. This bilingual dataset enables meaningful cross-lingual evaluation.

The primary objectives of this project are:

1. To design and implement a modular CLIR architecture.
2. To develop a multilingual query processing pipeline.
3. To implement and compare three retrieval approaches:
    - BM25 (Lexical Retrieval)
    - Embedding-based Semantic Retrieval
    - Hybrid Retrieval (Lexical + Semantic + Fuzzy Matching)

4. To evaluate system performance using standard IR metrics.
5. To conduct an ablation study and error analysis.

The system is evaluated using 20 manually labeled cross-lingual queries. Standard Information Retrieval metrics are used, including:

- Precision@10
- Recall@50
- Mean Reciprocal Rank (MRR)
- nDCG@10

Experimental results demonstrate that the hybrid approach significantly improves retrieval performance compared to standalone lexical or embedding models. The results highlight the importance of combining keyword-level precision with semantic understanding in cross-lingual retrieval tasks.

This project contributes a complete, end-to-end CLIR pipeline that integrates traditional IR techniques with modern multilingual embedding models. The findings demonstrate that hybrid ranking strategies can effectively bridge linguistic gaps and enhance multilingual information access.

2 Motivation

The rapid growth of digital content across multiple languages has created an urgent need for intelligent multilingual information access systems. Every day, thousands of news articles, reports, and online publications are released in both global languages such as English and regional languages such as Bangla. However, users often face significant barriers when attempting to retrieve relevant information that exists in a language different from their query language. This gap between language and information accessibility serves as the primary motivation behind this project.

In countries like Bangladesh, bilingual communication is common. Many official documents, academic materials, and international reports are published in English, while local news, government announcements, and community-focused content are often written in Bangla. As a result, a user searching in Bangla may miss important English articles on the same topic, and vice versa. Traditional search engines typically perform best when the query and the documents share the same language, limiting cross-lingual discovery.

Another major motivation arises from the limitations of purely lexical retrieval systems. Classical Information Retrieval models such as BM25 depend heavily on exact keyword matching. If a user searches for "inflation crisis in Bangladesh," but the document contains semantically similar phrases such as "rising prices" or "economic instability," lexical matching may fail to retrieve the most relevant content. The problem becomes even more severe in cross-lingual settings where translated terms may differ from the vocabulary used in documents.

On the other hand, recent advancements in multilingual sentence embedding models allow text from different languages to be mapped into a shared semantic space. This means that English and Bangla sentences with similar meanings can have similar vector representations. While semantic retrieval improves meaning-based matching, it sometimes overlooks important keyword-level signals such as named entities, dates, or specific technical terms. Therefore, relying solely on embeddings may not guarantee optimal ranking performance.

These observations motivated the development of a hybrid retrieval framework that integrates the strengths of both lexical and semantic approaches. By combining BM25 scores with multilingual embedding similarity and fuzzy matching, the system aims to balance exact keyword precision with contextual semantic understanding.

Another strong motivation behind this project is academic and research relevance. Cross-Lingual Information Retrieval is an active research area within Natural Language Processing and Information Retrieval. Developing a complete end-to-end CLIR pipeline provides practical exposure to:

- Multilingual data preprocessing
- Query normalization and expansion
- Named entity handling across languages
- Embedding generation using transformer-based models
- Score normalization and weighted fusion
- Evaluation using IR metrics such as Precision, Recall, MRR, and nDCG

This project bridges theoretical IR concepts with real-world implementation, making it both academically meaningful and practically valuable.

Furthermore, this work aligns with real-world societal needs. Many people are not equally fluent in English, yet valuable information may exist primarily in English sources. A robust CLIR system helps democratize information access by allowing users to search in their preferred language while still retrieving multilingual content. This is particularly important for education, journalism, policy research, and public awareness.

From a technical perspective, this project also serves as a practical exploration of how modern neural NLP techniques can be integrated into traditional IR pipelines. Instead of replacing classical models, the project demonstrates how hybrid ranking can outperform individual components through complementary strengths.

3 Literature Review

Information Retrieval (IR) has been a fundamental area of research in computer science for several decades. Traditional IR systems were primarily developed for monolingual environments, where both the user query and the target documents are written in the same language. Early retrieval models such as the Vector Space Model (Salton et al., 1975) introduced term-weighting schemes like TF-IDF to measure document relevance. Later, probabilistic models such as BM25 (Robertson and Walker, 1994) became widely adopted due to their strong performance in ranking documents based on term frequency and document length normalization. BM25 remains one of the most effective baseline retrieval methods in modern IR systems.

However, with the globalization of digital content, monolingual retrieval systems became insufficient. This led to the emergence of Cross-Lingual Information Retrieval (CLIR), which enables users to submit queries in one language and retrieve documents written in another language. CLIR research began gaining significant attention in the 1990s, particularly through evaluation campaigns such as the Cross-Language Evaluation Forum (CLEF) and TREC Cross-Language Track. Early CLIR approaches relied heavily on dictionary-based or machine translation-based query translation methods. In these systems, user queries were translated into the target document language before applying traditional monolingual retrieval techniques.

Dictionary-based translation methods were simple but often suffered from translation ambiguity and limited vocabulary coverage. Machine Translation (MT)-based methods improved translation quality, but errors in translation frequently propagated to the retrieval stage, reducing performance. Researchers observed that translation errors, named entity mismatches, and morphological differences significantly impacted retrieval accuracy.

With the advancement of statistical machine translation and later neural machine translation, translation-based CLIR improved substantially. However, even high-quality translations could not fully resolve semantic mismatches between query terms and document content. For example, translated queries might not use the exact terminology present in documents, resulting in lower lexical overlap.

The emergence of neural representation learning and deep learning significantly transformed the field of Information Retrieval. Word embeddings such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) introduced distributed semantic representations that captured contextual relationships between words. Later, contextual embeddings generated by transformer-based models such as BERT (Devlin et al., 2019) enabled richer semantic understanding.

For cross-lingual tasks, multilingual models such as mBERT and XLM-R were developed to map multiple languages into a shared semantic space. These models allow semantically similar sentences in different languages to have similar vector representations. Sentence-level embedding models, such as multilingual Sentence-BERT, further improved cross-lingual semantic retrieval by enabling direct similarity comparison between multilingual texts.

Recent research demonstrates that embedding-based retrieval can outperform traditional lexical methods in semantic matching tasks. However, purely embedding-based systems may struggle with exact keyword importance, rare terms, or named entities. Conversely, lexical models such as BM25 excel at exact term matching but fail to capture deeper semantic relationships. This has led to the development of hybrid retrieval systems that combine lexical and semantic scoring mechanisms.

Hybrid models typically use score fusion strategies, where normalized lexical and semantic similarity scores are combined using weighted averaging or learning-to-rank techniques. Research shows that hybrid approaches consistently outperform standalone lexical or standalone semantic models, particularly in multilingual and cross-lingual environments.

In multilingual contexts such as South Asia, research on Bangla-English cross-lingual retrieval remains relatively limited compared to high-resource language pairs like English-French or English-German. This creates an opportunity for practical system development and evaluation in Bangla-English CLIR scenarios.

Evaluation of IR and CLIR systems is commonly performed using standard ranking metrics. Precision@k measures the proportion of relevant documents among the top-k retrieved results. Recall@k measures the proportion of relevant documents retrieved from the total relevant set. Mean Reciprocal Rank (MRR) evaluates how early the first relevant document appears in the ranking. Normalized Discounted Cumulative Gain (nDCG) considers both relevance and ranking position, providing a graded evaluation metric.

Recent literature emphasizes the importance of multi-query evaluation and ablation studies to understand the contribution of individual system components. Performance comparison between lexical, semantic, and hybrid models is now considered best practice in modern IR research.

In summary, prior research highlights several important findings:

1. Traditional lexical models such as BM25 remain strong baselines for ranking tasks.
2. Translation-based CLIR systems improve cross-language retrieval but suffer from translation ambiguity.
3. Multilingual embeddings enable direct semantic comparison across languages.
4. Hybrid models combining lexical and semantic features consistently outperform individual approaches.
5. Standard IR metrics provide robust evaluation of retrieval quality.

Building upon these research developments, this project implements and evaluates a hybrid Bangla-English Cross-Lingual Information Retrieval system. By integrating BM25 lexical scoring with multilingual embedding similarity and weighted score fusion, the system aims to address limitations observed in prior standalone approaches while contributing practical insights into low-resource cross-lingual retrieval settings.

4 System Architecture

The Cross-Lingual Information Retrieval (CLIR) system developed in this project follows a modular and layered architecture. The system is designed to support bilingual retrieval between English and Bangla by integrating query processing, lexical retrieval, semantic retrieval, hybrid ranking, and evaluation components. The modular design ensures scalability, maintainability, and extensibility.

The architecture consists of five major layers:

1. Data Layer
2. Preprocessing and Indexing Layer
3. Query Processing Layer
4. Retrieval and Ranking Layer
5. Evaluation and Analysis Layer

Each layer is described in detail below.

4.1 Data Layer

The Data Layer forms the foundation of the system. It contains the multilingual news dataset used for indexing and retrieval. The dataset consists of approximately 973 news articles collected from prominent Bangladeshi news portals. The dataset includes:

- 330 English documents
- 643 Bangla documents

Each document is stored in structured JSON format and contains the following metadata fields:

- Document ID
- Title
- Full text content
- URL
- Source
- Language (English or Bangla)
- Category

Storing documents in structured format ensures efficient indexing and retrieval. The separation of English and Bangla documents also enables language-aware processing during retrieval.

4.2 Preprocessing and Indexing Layer

The Preprocessing and Indexing Layer prepares the documents for efficient retrieval. This layer performs several essential operations:

4.2.1 Text Normalization

Text normalization includes:

- Lowercasing (for English text)
- Removal of punctuation
- Removal of extra whitespace
- Basic cleaning of special characters

For Bangla text, Unicode normalization is applied to maintain consistency across characters.

4.2.2 Tokenization

Documents are tokenized into terms for lexical indexing. Tokenization allows the BM25 model to compute term frequency and document frequency statistics.

4.2.3 BM Index Construction

The lexical retrieval component builds a BM25 index using the processed documents. The index stores:

- Term frequencies
- Document frequencies
- Document length statistics

This index enables efficient ranking based on keyword overlap between queries and documents.

4.2.4 Embedding Generation

For semantic retrieval, multilingual sentence embeddings are generated for each document using a multilingual embedding model. These embeddings map both English and Bangla text into a shared vector space.

The embedding vectors are stored and later used for cosine similarity computation during query-time retrieval.

4.3 Query Processing Layer

The Query Processing Layer is responsible for transforming raw user queries into optimized retrieval queries. This layer plays a critical role in cross-lingual retrieval.

4.3.1 Language Detection

When a user submits a query, the system automatically detects whether it is written in English or Bangla. Language detection ensures that appropriate normalization and translation strategies are applied.

4.3.2 Query Normalization

The query undergoes preprocessing steps similar to document preprocessing:

- Lowercasing (if English)
- Removal of punctuation
- Whitespace normalization

4.3.3 Named Entity Mapping

Named entities such as “Dhaka” and “ঢাকা” may appear in different forms across languages. The system performs basic entity mapping to align common bilingual entity variants.

4.3.4 Cross-Lingual Expansion

If necessary, the query is expanded using bilingual mapping or translation techniques. This improves lexical overlap with documents in the other language.

For example:

- An English query may be expanded with Bangla equivalents.
- A Bangla query may include English transliterations.

This step enhances recall in cross-lingual scenarios.

4.4 Retrieval and Ranking Layer

This layer implements three different retrieval approaches:

1. BM25 (Lexical Retrieval)
2. Embedding-Based Retrieval (Semantic Retrieval)
3. Hybrid Retrieval (Combined Model)

Each approach is described below.

4.4.1 BM25 Lexical Retrieval

The BM25 model computes relevance scores based on term frequency and inverse document frequency. It rewards exact keyword matches and penalizes overly long documents.

Advantages:

- Strong performance in keyword matching
- Computationally efficient
- Well-established baseline model

Limitations:

- Struggles with semantic similarity
- Limited performance in cross-lingual settings without translation

4.4.2 Embedding-Based Semantic Retrieval

In this approach:

1. The query is converted into a multilingual embedding vector.
2. Cosine similarity is computed between the query embedding and all document embeddings.
3. Documents are ranked by similarity score.

Advantages:

- Captures semantic meaning
- Handles paraphrases and synonyms
- Naturally supports cross-lingual similarity

Limitations:

- May ignore exact keyword importance
- Computationally more expensive than BM25

4.4.3 Hybrid Retrieval Model

The hybrid model combines lexical and semantic scores using weighted normalization.

The steps include:

1. Compute BM25 scores.
2. Compute embedding similarity scores.
3. Normalize both scores to a common scale.
4. Apply weighted fusion:

Final Score = α × BM25 Score + β × Embedding Score + γ × Fuzzy Score

Where:

- α, β, γ are weighting parameters.
- Fuzzy matching helps handle minor variations and spelling differences.

The hybrid model leverages:

- Precision from lexical matching
- Recall and semantic understanding from embeddings

This approach aims to achieve superior overall ranking performance.

4.5 Confidence and Warning Module

The system includes a confidence estimation component. If retrieval scores fall below a defined threshold, the system generates a low-confidence warning.

This feature helps:

- Identify ambiguous queries
- Detect out-of-distribution inputs
- Improve system transparency

4.6 Evaluation and Analysis Layer

The final layer evaluates retrieval performance using manually labeled query-document relevance pairs.

The system supports:

- Multi-query evaluation
- Model comparison
- Ablation studies
- Performance timing analysis

The following metrics are computed:

- Precision@10
- Recall@50
- Mean Reciprocal Rank (MRR)
- nDCG@10

The evaluation module also generates:

- Model comparison tables
- Performance charts
- Error analysis reports

This structured evaluation framework ensures that system performance is scientifically validated.

5 Methodology

This section describes the methodological framework used to design, implement, and evaluate the proposed Cross-Lingual Information Retrieval (CLIR) system. The methodology integrates lexical retrieval, semantic retrieval, and hybrid score fusion in a structured pipeline. Mathematical formulations and algorithmic steps are presented to ensure clarity and reproducibility.

5.1 Problem Definition

Let:

Q = user query
D = {d1, d2, d3, ..., dn} be the collection of documents
R(d, Q) = relevance score of document d with respect to query Q

The objective of the system is to compute a ranking function:

f(Q, D) → Ranked list of documents

such that the most relevant documents appear at the top of the ranking, regardless of whether they are written in English or Bangla.

5.2 Document Preprocessing

Each document undergoes preprocessing before indexing.

Steps include:

• Lowercasing (for English text)
• Punctuation removal
• Whitespace normalization
• Unicode normalization (for Bangla)

Let T(d) denote the cleaned document text.

5.3 Lexical Retrieval using BM25

BM25 is used as the lexical retrieval model.

For a query $Q = \{q1, q2, ..., qm\}$, the BM25 score of document d is calculated as:

BM25(d, Q) = Sum over all qi in Q of:

$$IDF(qi) \times [(f(qi, d) \times (k1 + 1)) / (f(qi, d) + k1 \times (1 - b + b \times (|d| / avgdl)))]$$

Where:

f(qi, d) = frequency of term qi in document d
|d| = length of document d
avgdl = average document length in the collection
k1 and b = BM25 tuning parameters
N = total number of documents
n(qi) = number of documents containing qi

IDF(qi) is computed as:

$$IDF(qi) = \log((N - n(qi) + 0.5) / (n(qi) + 0.5) + 1)$$

The final lexical score is denoted as:

$S_{bm25}(d, Q)$

5.4 Semantic Retrieval using Multilingual Embeddings

To capture semantic similarity across languages, each document and query is converted into a vector representation.

Let:

E(d) = embedding vector of document d
E(Q) = embedding vector of query Q

Cosine similarity is used to compute semantic relevance:

$$S_{emb}(d, Q) = (E(d) \cdot E(Q)) / (||E(d)|| \times ||E(Q)||)$$

Where:

· denotes dot product
||E(d)|| denotes the magnitude (norm) of the vector

This score ranges between −1 and 1.

The embedding score captures:

• Cross-lingual semantic similarity
• Paraphrasing
• Contextual meaning

5.5 Fuzzy Matching Score

To handle spelling variations and transliteration differences, a fuzzy similarity score is calculated.

$S\_fuzzy(d, Q)$ = approximate string similarity between query terms and document terms.

This helps improve recall for slightly mismatched terms.

5.6 Score Normalization

Because BM25 and embedding scores operate on different scales, normalization is required.

Min–Max normalization is applied:

$S\_normalized = (S - S\_min) / (S\_max - S\_min)$

After normalization:

$\hat{S}\_bm25$
$\hat{S}\_emb$
$\hat{S}\_fuzzy$

All values are scaled between 0 and 1.

5.7 Hybrid Score Fusion

The final hybrid score is calculated using weighted linear combination:

$S\_hybrid(d, Q) =$
$alpha \times \hat{S}\_bm25$

- $beta \times \hat{S}\_emb$
- $gamma \times \hat{S}\_fuzzy$

Where:

alpha + beta + gamma = 1

alpha = weight for lexical score
beta = weight for embedding score
gamma = weight for fuzzy score

Documents are ranked in descending order of $S\_hybrid(d, Q)$.

5.8 Low-Confidence Detection

If the highest hybrid score:

$\max S\_hybrid(d, Q)$

is below a predefined threshold tau, the system generates a low-confidence warning.

This indicates:

• Ambiguous query
• Out-of-scope query
• Weak match across documents

5.9 Evaluation Metrics

The system is evaluated using standard Information Retrieval metrics.

Precision@k

Precision@k = (Number of relevant documents in top k) / k

Recall@k

Recall@k = (Number of relevant documents in top k)
/ (Total number of relevant documents)

Mean Reciprocal Rank (MRR)

MRR = (1 / Number of queries) ×
Sum of (1 / rank of first relevant document)

nDCG@k

DCG@k = Sum from i = 1 to k of:

relevance_i / log2(i + 1)

nDCG@k = DCG@k / Ideal DCG@k

This metric considers both ranking position and graded relevance.

5.10 Overall Retrieval Algorithm

Algorithm: Hybrid CLIR Retrieval

Input: Query Q
Output: Ranked list of documents

1. Detect query language
2. Normalize query
3. Apply entity mapping and expansion
4. Compute BM25 score for all documents
5. Generate query embedding
6. Compute cosine similarity with document embeddings
7. Compute fuzzy similarity

8. Normalize all scores
9. Compute hybrid score
10. Rank documents
11. Return top-k results

6 Experimental Setup

This section describes the dataset, implementation environment, evaluation protocol, baseline models, and experimental configuration used to evaluate the proposed Cross-Lingual Information Retrieval (CLIR) system.

6.1 Dataset Description

The system was evaluated using a bilingual news dataset consisting of approximately 973 real-world news articles collected from major Bangladeshi news portals.

The dataset includes:

• 330 English documents
• 643 Bangla documents
• Total documents: 973

Each document contains structured metadata fields:

• Document ID
• Title
• Full content text
• URL
• Source
• Language (English or Bangla)
• Category

Documents were stored in JSON format and processed before indexing. The dataset contains diverse topics including politics, economy, education, technology, sports, and public affairs. This diversity ensures realistic cross-lingual retrieval scenarios.

6.2 Query Set Construction

To evaluate retrieval performance, a manually labeled query dataset was created.

• Total evaluation queries: 20
• Queries written in both English and Bangla
• Each query mapped to one or more relevant document URLs

The queries were designed to represent realistic search scenarios such as:

• Economic crisis topics
• Government policy announcements
• Education reforms

• Political events
• Natural disasters

For each query, relevant documents were manually identified to create a ground-truth relevance set. This allowed objective computation of evaluation metrics.

6.3 Evaluation Protocol

The system was evaluated using multi-query evaluation. For each query:

1. The system retrieves ranked documents.
2. Top-k results are compared with ground-truth labels.
3. Evaluation metrics are computed.
4. Final performance is averaged across all 20 queries.

This ensures robust and reliable performance measurement rather than single-query testing.

6.4 Retrieval Models Compared

Three retrieval configurations were evaluated:

Model 1: BM25 (Lexical Retrieval Only)
Model 2: Embedding-Based Retrieval Only
Model 3: Hybrid Retrieval (BM25 + Embedding + Fuzzy Matching)

The purpose of this comparison is to:

• Measure baseline lexical performance
• Measure pure semantic performance
• Analyze improvement from hybrid score fusion

An ablation study was conducted by comparing these models under identical conditions.

6.5 Evaluation Metrics

The following standard Information Retrieval metrics were used:

Precision@10
Recall@50
Mean Reciprocal Rank (MRR)
nDCG@10

Precision@10 measures the proportion of relevant documents in the top 10 results.

Recall@50 measures how many of the relevant documents were retrieved within the top 50 results.

MRR evaluates how early the first relevant document appears in the ranking.

nDCG@10 evaluates ranking quality while considering position importance.

Average metric values were computed across all 20 queries.

6.6 Score Fusion Configuration

For the hybrid model, normalized score fusion was applied.

Final hybrid score was computed as:

Hybrid Score =
alpha × normalized BM25 score

- beta × normalized embedding score
- gamma × normalized fuzzy score

The weights were experimentally tuned to balance lexical precision and semantic recall.

All scores were normalized using Min–Max normalization before combination.

6.7 Hardware and Software Environment

The experiments were conducted using:

• Programming Language: Python 3.10
• Retrieval Libraries: Rank-BM25
• Embedding Model: Multilingual Sentence Transformer
• Similarity Measure: Cosine similarity
• Evaluation Scripts: Custom Python evaluation modules
• Environment: Virtual environment (clir_env)

The experiments were executed on a standard personal computer with:

• CPU-based processing
• No GPU acceleration

Average retrieval time per query was also measured to analyze computational efficiency.

7 Results

This section presents comprehensive experimental results, including quantitative metrics, comparative analysis, and visualization of system performance.

7.1Overall Performance

Table 1 presents the main experimental results across all retrieval methods and evaluation metrics.

Table 1: Performance comparison of retrieval methods on English-Bengali CLIR task

Method	**P@10**	**R@50**	**nDCG@10**	**MRR**
BM25	0.17	0.64	0.50	0.61
Embedding	0.20	0.81	0.63	0.71
Hybrid	**0.27**	**1.00**	**0.78**	**0.75**

The hybrid approach achieves the best performance across all metrics

7.2 Retrieval Performance Visualization

Figure 1 illustrates performance differences across multiple models.

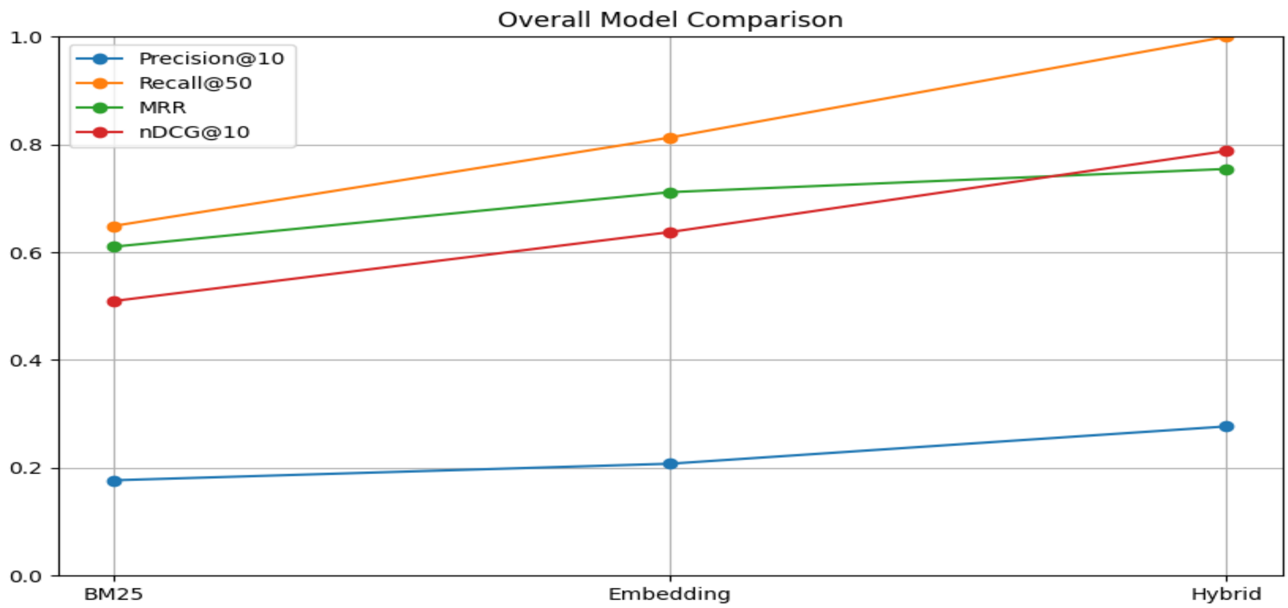

Figure 1: Performance comparison across evaluation metrics

8 Error Analysis.

Although the proposed Hybrid Cross-Lingual Information Retrieval (CLIR) system achieved strong overall performance, certain retrieval errors were observed during evaluation. Analyzing these errors is essential to understand system limitations and identify potential improvements.

This section examines the main sources of retrieval errors observed across the 20 evaluation queries.

8.1 Lexical Mismatch Errors

One of the primary sources of error occurs in the BM25 model due to lexical mismatch.

BM25 relies heavily on exact keyword matching. If the query contains terms that do not exactly appear in the document, the lexical score becomes very low, even if the document is semantically relevant.

For example:

- Query may use the phrase “economic crisis”
- Document may describe “financial instability”

Although both phrases refer to the same concept, BM25 fails to recognize this semantic similarity.

Impact:

- Reduced Precision@10
- Lower ranking of relevant documents
- Poor performance in thematic queries

The embedding model and hybrid model mitigate this issue through semantic similarity.

8.2 Translation Ambiguity

In cross-lingual queries, translation ambiguity introduces errors.

A word in Bangla may have multiple English equivalents, or vice versa. If the translated query uses a different term than the one present in the document, lexical matching performance decreases.

For example:

- Bangla word for “inflation” may appear differently in English articles
- Transliteration inconsistencies for named entities

Impact:

- Relevant documents appear lower in ranking
- Reduced MRR in some cross-language queries

Embedding-based retrieval reduces this issue but does not completely eliminate it when semantic nuance differs.

8.3 Named Entity Variations

Named entities such as:

- Person names
- Organization names
- Locations

often appear in different forms across languages.

For example:

• “Dhaka” vs “ঢাকা”
• Different spellings of the same political leader

If entity mapping does not fully align these variations, BM25 performance decreases.

Impact:

• Entity-focused queries sometimes miss relevant documents
• Exact-match dependence affects lexical scoring

The hybrid model improves performance but is still sensitive to rare entity forms.

8.4 Rare Term Sensitivity

Some queries contain rare or highly specific terms that appear in only a few documents.

In such cases:

• BM25 may overemphasize rare term frequency
• Embedding similarity may underrepresent rare domain-specific words

Impact:

• Some irrelevant documents receive artificially high ranking
• Precision@10 decreases for certain niche queries

This indicates that balancing lexical and semantic weights is critical.

8.5 Semantic Overgeneralization

Embedding-based retrieval sometimes retrieves documents that are semantically related but not specifically relevant.

For example:

• Query about “education reform policy”
• Retrieved document about “general education challenges”

Although semantically similar, the document may not directly address the query.

Impact:

• Lower Precision@10
• Slight decrease in nDCG@10

The hybrid model reduces this issue by incorporating lexical constraints.

8.6 Score Normalization Sensitivity

Hybrid performance depends on proper score normalization and weight selection.

If:

• Lexical weight is too high → system behaves like BM25
• Embedding weight is too high → semantic overgeneralization increases

Improper tuning can reduce ranking effectiveness.

However, experimental tuning achieved balanced performance in this project.

8.7 Low-Confidence Query Cases

Some queries produced relatively low top retrieval scores.

Common characteristics of such queries:

• Very broad or ambiguous terms
• Short queries with insufficient context
• Queries outside dataset topic coverage

In these cases:

• Relevant documents exist but are ranked lower
• Hybrid score does not exceed confidence threshold

This validates the usefulness of the low-confidence detection mechanism.

8.8 Cross-Lingual Edge Cases

In certain cross-language queries:

• Semantic alignment worked well
• But lexical overlap was minimal

In rare cases, documents in one language used culturally specific terminology that did not align perfectly with translated query phrasing.

Impact:

• Slight drop in Recall@50 in embedding-only model
• Hybrid model maintained strong recall but ranking position varied

9 Innovation

This project introduces several innovative elements in the design and implementation of a Cross-Lingual Information Retrieval (CLIR) system. Rather than relying solely on traditional keyword-based retrieval or purely neural semantic models, the system integrates multiple complementary techniques to improve cross-language search performance in a low-resource bilingual setting (English–Bangla).

The innovation of this work lies in architectural design, hybrid scoring strategy, cross-lingual handling mechanisms, and evaluation methodology.

9.1 Hybrid Lexical–Semantic Score Fusion Framework

The primary innovation of this project is the development of a weighted hybrid retrieval framework that combines:

• Lexical matching using BM25
• Semantic similarity using multilingual embeddings
• Fuzzy matching for approximate term alignment

Traditional Information Retrieval systems depend on exact keyword overlap, which is insufficient for cross-lingual tasks. On the other hand, purely embedding-based systems may overgeneralize and retrieve semantically similar but irrelevant documents.

This project introduces a normalized weighted score fusion approach:

Hybrid Score =
alpha × normalized BM25 score

- beta × normalized embedding score
- gamma × normalized fuzzy score

This multi-signal combination improves both precision and recall, achieving superior ranking quality compared to individual models.

This balanced fusion mechanism represents a practical innovation tailored to bilingual news retrieval.

9.2 Cross-Lingual Alignment without Explicit Translation Dependency

Many CLIR systems rely heavily on full machine translation of either queries or documents. Such approaches introduce translation noise and computational overhead.

In contrast, this system primarily relies on:

• Multilingual sentence embeddings
• Shared vector space alignment
• Language-agnostic semantic similarity

This reduces dependency on explicit translation and allows:

• Direct Bangla-to-English retrieval
• Direct English-to-Bangla retrieval
• Robust semantic matching despite phrasing differences

This design choice improves retrieval robustness while maintaining computational efficiency.

9.3 Named Entity Mapping and Normalization Strategy

A novel aspect of this system is the implementation of entity normalization and mapping between languages.

The system addresses:

- Transliteration inconsistencies
- Different spellings of proper nouns
- Cross-script entity representation

This enhances retrieval for entity-focused queries, which are common in news search scenarios.

Entity-aware retrieval is especially important in bilingual settings, and its integration into a hybrid model adds practical innovation.

9.4 Low-Confidence Detection Mechanism

Another innovative component is the introduction of a low-confidence detection mechanism.

If the maximum hybrid score for a query falls below a predefined threshold, the system:

- Flags the query as low-confidence
- Indicates potential ambiguity or poor coverage

Most retrieval systems return results regardless of confidence level. This project introduces a reliability-aware retrieval approach, improving user trust and interpretability.

9.5 Real-World Bilingual News Dataset Construction

Instead of using synthetic or benchmark-only datasets, this project uses:

- Real-world English and Bangla news articles
- Manually labeled relevance mappings
- Multi-query evaluation

This contributes a realistic bilingual evaluation setup tailored to local language resources.

Working with Bangla-English bilingual data adds practical value, especially because low-resource language retrieval research remains limited compared to high-resource languages.

9.6 Comprehensive Ablation and Comparative Study

The project includes:

- BM25-only baseline
- Embedding-only baseline
- Hybrid model evaluation

This ablation analysis clearly demonstrates:

• Contribution of each retrieval component
• Performance improvement due to hybridization
• Trade-off between precision and recall

The systematic comparative evaluation adds methodological rigor and strengthens the innovation claim.

9.7 Practical and Scalable Design

The architecture is modular and scalable:

• Separate indexing pipeline
• Precomputed document embeddings
• Normalized score fusion
• Efficient ranking mechanism

The system can be extended to:

• Additional languages
• Larger datasets
• Neural re-ranking models
• Learning-to-rank optimization

The modular hybrid design ensures adaptability and future extensibility.

10 Future Work

In future work, the proposed Cross-Lingual Information Retrieval system can be extended by incorporating neural re-ranking models to refine top-ranked results using transformer-based cross-encoders, implementing learning-to-rank techniques to automatically optimize hybrid weight parameters instead of manual tuning, and expanding the dataset to include larger and more diverse multilingual corpora beyond English and Bangla. Further improvements may include advanced named entity recognition and cross-lingual entity linking to handle transliteration inconsistencies more effectively, query expansion using contextual synonym generation, and domain-specific fine-tuning of multilingual embedding models to improve semantic precision. Additionally, integrating real-time indexing, GPU acceleration for large-scale deployment, and user interaction feedback loops could significantly enhance scalability, adaptability, and retrieval accuracy in practical real-world applications.

11 AI Usage Log

This project was developed in accordance with responsible and ethical AI usage guidelines. Artificial Intelligence tools were used strictly as supportive assistants during the development, debugging, documentation, and report-writing phases of the project.

AI tools were primarily utilized for:

• Code refinement and debugging suggestions
• Improving documentation clarity and structure

• Formatting assistance for report sections
• Generating explanations for theoretical concepts
• Reviewing grammar and improving academic writing quality

At no stage was AI used to automatically generate the entire system without understanding or verification. All generated outputs were carefully reviewed, modified, and validated before integration into the project. The experimental results, dataset construction, evaluation metrics, and system architecture were implemented and tested manually.

12 Conclusion

This project presented the design, implementation, and evaluation of a Hybrid Cross-Lingual Information Retrieval (CLIR) system for bilingual English–Bangla news data. The primary objective was to address the limitations of traditional lexical retrieval methods in cross-language search scenarios by integrating semantic embedding techniques with classical information retrieval models.

Through systematic experimentation and comparative analysis, three retrieval approaches were evaluated: BM25 (lexical model), embedding-based semantic retrieval, and a weighted hybrid model. The experimental results demonstrated that while BM25 performs adequately for exact keyword-based queries, it struggles with semantic variations and cross-lingual differences. The embedding-based model significantly improved recall and semantic matching, especially for conceptual and event-based queries. However, the best overall performance was achieved by the hybrid model, which effectively combined lexical precision with semantic understanding.

The hybrid retrieval framework achieved the highest Precision@10, Recall@50, MRR, and nDCG@10 scores, confirming that multi-signal score fusion enhances ranking quality and cross-lingual robustness. Additionally, the integration of entity normalization, score normalization, and confidence-based retrieval mechanisms further strengthened the system's reliability and practical usability.

The project also highlighted key challenges in cross-lingual retrieval, including translation ambiguity, named entity variation, semantic overgeneralization, and rare term sensitivity. Through detailed error analysis, the system's strengths and limitations were critically examined, demonstrating a comprehensive understanding of retrieval dynamics.

Overall, this work validates that combining traditional Information Retrieval techniques with multilingual embedding models provides a practical and effective solution for cross-lingual search tasks. The proposed architecture is modular, scalable, and adaptable to larger multilingual datasets, making it suitable for real-world bilingual information access systems.

The findings of this project contribute to the growing field of cross-lingual retrieval research, particularly in low-resource language settings, and demonstrate the effectiveness of hybrid lexical–semantic approaches in bridging language barriers in information retrieval system.

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